

Transfer

One word, sixteen senses, and the two a security room reaches for first

Contents

1	Why this entry	2
1.1	How the entry was built	2
2	The senses	2
2.1	Group I. What moves is a policy or a model	3
2.2	Group II. What moves is a claim, a method, or a person	5
2.3	Group III. Homonyms	6
2.4	Group IV. Two senses your corpus never uses	7
3	The census	7
4	The blind pass	9
5	Where two of your files disagree	10
6	What to say instead	10

1 Why this entry

Question 39 says what this file is for.

“Try to match terminologies about opponent modeling, transfer learning, cyber environments across cybersecurity, AI, and human factors community.”

q_n_a.tex, question 39

The same want turns up in the middle of the rebuttal, where it is not a project idea but a thing blocking the next paragraph.

“In general I see the word generalization and transferable used for multiple meanings. I wonder if there are terms that can disentangle the various meanings this word served in the various comments. For example, we can clarify the terms first, and then address concerns about each of them later.”

author note T20, [Aug25th]_Rebuttal

“I suppose for security people the term generalization just means the capability to counteract different attackers in general? Or, what is the most cybersecurity relevant definition of this term? One possibility I can think of, at least for Reviewer B, is to be able to handle unseen attackers. What is the term for that? Zero shot transfer?”

same note

Your own publication list already carries the word in two senses. *Accounting for transfer of learning using human behavior models*, with Malloy, Fang and Gonzalez at HCOMP, is about a person carrying a skill from one task to the next. *Crossing the Cyber Divide: Sim-to-Sim and Sim-to-Real Transfer for RL Agents*, with Saika and Piplai at RAISE, is about a policy being executed in a simulator it was never trained in. Neither title tells a reader which one is meant, and the two do not share a single technical commitment.

1.1 How the entry was built

A census, not a sample. Every occurrence of **transfer** and its morphological variants was enumerated across seven parts of your own work before any sense was written down: the website, the FOE-Dreamer manuscript, the nine RAISE section files, the Learn Structure sections, the DesignTheGame core idea, the PickYourBattles mind map, and the August rebuttal. That is 59 files and 267 tokens. The sense inventory was fixed and frozen first, aligned to outside sources that define the term where such a source exists, and only then was the corpus assigned against it. Section 4 reports the blind pass.

Tiers are the ones on the Tech Fluency page. Active senses get all four fields. Working senses get a definition and a discrimination. Peripheral senses get a definition and a pointer.

2 The senses

2.1 Group I. What moves is a policy or a model

P1. Policy transfer across environments

active

Definition. Executing a policy trained in one environment inside a second environment whose state space, action space, or dynamics differ from the first, with the source policy left unchanged.

“We view sim-to-sim and sim-to-real transfer as instances of the same underlying problem: aligning policies across environments with different state and action representations.”

RAISE, 1_introduction.tex

Distinguishes it from. P6, generalization. In P1 the target is a different program with its own vocabulary, so a translator has to be built before the policy can act at all. In P6 the target is another configuration of the same program and no translator exists or is needed.

Operational test. Is there a second environment with its own state encoding and its own action names? If a wrapper has to be written before the policy can emit a legal action, it is P1.

In my project. The whole of the RAISE paper. 112 of the 267 tokens in the census sit in its nine section files, and every one of them is P1 or P2.

P2. Sim-to-sim and sim-to-real

active

Definition. A policy transfer in which the target is the place the policy is meant to run, or a stand-in for it, rather than another simulator of equal standing. Your position is that the two are one problem with one difference.

“Sim-to-real is the same problem as sim-to-sim, with one extra difficulty: the target is only partially known.”

website, transfer-to-realistic-environments.md

“Most RL defenders learn inside a simulator and then transfer the policy to the network they protect.”

FOE-Dreamer main.tex, Introduction

Distinguishes it from. P1 by the target only. Supplying the missing ground truth about the target collapses P2 back into P1, which is still hard, because finding the abstraction that relates two fully specified environments is itself computationally difficult.

Operational test. Is the target the deployment, or a thing whose only job is to resemble the deployment? If the answer is yes, the uncertainty about the target is part of the problem statement and not an experimental inconvenience.

In my project. Two of your projects use this sense against each other. FOE-Dreamer removes the simulator stage so that no P2 step is needed. The RAISE work studies the P2 step directly. The website states the first position in one line.

“If transfer from a simulator is where the fidelity is lost, train where the fidelity is.”

website, training-in-realistic-environments.md

P3. Transfer to unseen opponents

active

Definition. A defender trained against one set of attacker classes holding up against an attacker

class it never saw, with the environment held fixed. What changes is the adversary, not the network.

Reviewer B *“Generalization is evaluated only on two fixed scripted attacker profiles and a single small topology, so it is unclear whether the opponent model learns transferable behavior or simply distinguishes two known scripts.”*

comment B6, second half

Distinguishes it from. P1 and P2, where the environment is what changes. This is the sense a security reader supplies by default when nobody names one, and it is the one the rebuttal had to separate out by hand.

Operational test. Name the thing that is new at test time. If it is an adversary rather than a network, it is P3.

In my project. Section 5.1 of the rebuttal splits three questions travelling under one word: holding up under variation inside the trained attacker classes, transfer to unseen attacker classes, and an adversary that adapts against the deployed defender. The paper claims the first. All 19 tokens in corpus G are P3, and none of them is P1.

P4. Transfer learning, in the survey sense

working

Definition. Pan and Yang, whose Definition 1 is the field’s standard statement: given a source domain $\mathcal{D}_S$ and learning task $\mathcal{T}_S$, a target domain $\mathcal{D}_T$ and learning task $\mathcal{T}_T$, transfer learning aims to help improve the learning of the target predictive function $f_T(\cdot)$ in $\mathcal{D}_T$ using the knowledge in $\mathcal{D}_S$ and $\mathcal{T}_S$, where $\mathcal{D}_S \neq \mathcal{D}_T$ or $\mathcal{T}_S \neq \mathcal{T}_T$. A domain is a pair $\mathcal{D} = \{\mathcal{X}, P(X)\}$ and a task is a pair $\mathcal{T} = \{\mathcal{Y}, P(Y | X)\}$.

Distinguishes it from. P1 and P2, which are claims about executing a policy, not about improving a predictive function. The file you are reading is named for P4 and almost nothing in the corpus is P4. What the RAISE method actually does is closer to unsupervised domain adaptation: the encoder sees unlabeled observations from the target and no reward from it, which is Pan and Yang’s transductive setting, while the policy takes no gradient step in the target at all.

“Pretraining requires only random-policy observations from each simulator, with no policy trajectories or target-domain rewards.”

RAISE, 3_methodology.tex

P5. Reuse inside one training run

working

Definition. A shared network or a shared representation carrying across iterations, epochs, or members of a population inside a single training procedure.

“NeuPL represents all policies as a single shared conditional neural network, which enables transfer learning across PSRO epochs.”

PickYourBattles, psro_lit_notes.md

Distinguishes it from. P4, which requires the source and target to be different domains or different tasks. Whether a later PSRO epoch is a different task is a real question and not a settled one, which is why this cell ends up ambiguous in Section 4. Your own world model shares weights across devices, which is the same idea and is likewise not P4.

P6. Generalization to unseen configurations

working

Definition. One policy performing on instances of the same environment family it was never trained on, with no gradient step at test time. Kirk, Zhang, Grefenstette and Rocktaschel give the formalism the field now uses for this.

“Poor transfer | unseen-configuration performance retention”

website, arl-baa-0122, 30-method-candidates-and-metrics.md

Distinguishes it from. P1. That table row is the clearest case in the corpus of the two being run together: the phrase says transfer and the metric beside it measures generalization. A metric column is a good place to catch this, because the metric has to name the thing that varies.

2.2 Group II. What moves is a claim, a method, or a person

C1. Method transfer between literatures

working

Definition. A measure, a construction, or a proof carried out of the field that produced it and into yours, with its presuppositions rechecked in the new setting.

“transfer blocked, with the blocker named in the measure’s own presuppositions”

Learn Structure, adapting.tex, table legend

“The transferable content is the evidence standard”

Learn Structure, taxonomy_v2.tex

Distinguishes it from. A citation. A citation moves a sentence. C1 moves an operation together with the conditions under which it is licensed, and the entire point of the Learn Structure adaptation section is that those conditions are what fails first. One of its five rows records a measure whose transfer is stopped by a presupposition stated in its own source.

C2. Whether a result survives the move

working

Definition. Whether a finding obtained in one environment remains true in another. The subject of the verb is a claim, not an agent.

“the ones you can train in fastest are the ones whose results transfer least”

website, cyber-environments-and-benchmarks/README.md

“Transfer is only evidence of deployment-readiness if it survives being actively doubted.”

website, toward-deployment/next.md

Distinguishes it from. P1 and P2, which succeed or fail as engineering. C2 can fail while P1 succeeds, and that case has a name on your own site: a transferred policy that scores well has either learned the task or learned the benchmark, and the score alone will not say which. A P1 success is therefore not C2 evidence until something separates the two.

C3. Transfer of learning, in a person

active

Definition. A person applying what was learned in one setting to another. Barnett and Ceci’s

taxonomy is the standard placement, distinguishing near from far along dimensions of content and of context, so that studies reaching opposite verdicts can be compared instead of averaged.

“General heuristics transfer badly.”

website, resources/research/think.md

“the heuristics did not transfer well, because each label such as “solve a simpler related problem” conceals many distinct sub-strategies that must themselves be learned”

DesignTheGame, on_mental_operations.tex

Distinguishes it from. Retention. A person who can recite an operation has retained it. Whether they deploy it on a problem that does not announce which operation it wants is C3, and the finding in the corpus is that the two come apart: people who can recite them still fail to deploy them, because knowing an operation and regulating when to use it are different capacities.

Operational test. Is the thing that moves inside a person? If the answer is yes, no amount of architecture talk applies and the evidence standard is behavioral.

In my project. All eight tokens in the DesignTheGame core idea are C3, as is the sixth phase of the capture-the-flag solve model, verification and transfer. This is also the sense of your HCOMP paper.

2.3 Group III. Homonyms

These share the spelling and nothing else. They are peripheral, and they are listed because three of them sit in the same folders as the senses above.

H1. Data transfer

peripheral

Bytes moving between hosts. It appears three times in the FOE-Dreamer manuscript, in the attack narrative and in the description of user activity: the intruder *dumps tables for outbound transfer*, and GHOSTS personas perform *file transfers between PCs*.

H2. Transfer entropy

peripheral

Schreiber’s conditional mutual information between a source process’s past and a target’s present. Nine of the eleven tokens in `taxonomy_v2.tex` are this, where it appears as one of the named alternatives that violate the axioms and is shown to assign zero bits to an arrow that must carry one.

H3. Transferable utility

peripheral

Cooperative game theory. Payoff can be moved between players by side payment, which is what licenses the Shapley construction. One token, in `taxonomy_reorganization_redo.tex`.

H4. State transfer, in control

peripheral

Driving a system from one state to another, priced by the input energy the controllability Gramian fixes. One token, in `adapting.tex`, inside the paragraph that replaces a binary controllability verdict with a cost.

H5. Institutional transfer

peripheral

A technology transfer grant, a transfer story told to a sponsor, a transfer student. Two tokens on the site and two in the mind map.

2.4 Group IV. Two senses your corpus never uses

This is the part worth carrying into a room. Both senses below are standard in cybersecurity, both are what a security audience may hear when you say the word without a qualifier, and neither occurs once in 267 tokens of your own writing.

S1. Attack transferability

active

Definition. An adversarial input crafted against one model succeeding against a second model that the attacker never had access to, which is what makes black-box attacks practical. Papernot, McDaniel and Goodfellow named the phenomenon and turned it into an attack procedure.

Distinguishes it from. Every sense above by the sign of the outcome. In P1 through C3 the moving is the goal and failure to move is the problem. In S1 the moving is the failure. A defense against S1 works by making transfer harder, which is the exact opposite of what the RAISE paper is optimizing.

Operational test. Does the party that benefits from the moving differ from the party that did the training? If yes, it is S1.

In my project. Nowhere, and that is the finding. A reader who arrives from adversarial machine learning will read the title *Sim-to-Sim and Sim-to-Real Transfer* against this sense first, and will expect a threat model.

S2. Risk transfer

active

Definition. A risk treatment option in which the financial consequence of a risk is moved to a third party, by insurance or by contract, rather than being reduced, avoided, or accepted. ISO/IEC 27005 carries it as one of the four standard options.

Distinguishes it from. Everything above by what moves. No model, no policy, no skill and no claim moves. A liability does, and the technical risk is unchanged after the move.

Operational test. Does someone pay? If the sentence can be completed with a premium, it is S2.

In my project. Nowhere. It is the sense a governance or risk-management audience will hear first, and it is the reason the phrase *transfer assumptions checklist* on the BeRealistic track page is worth renaming before that page reaches such an audience.

3 The census

One row per file. The sense column lists the senses present, not their counts, because a count per sense inside a mixed file cannot be checked from the table. Subtotals are token counts and they sum to 267.

File	Tokens	Senses
A. Open Book, content/, 26 files, 49 tokens		
toward-deployment/transfer-to-realistic-environments.md	7	P1, P2
toward-deployment/training-in-realistic-environments.md	5	P2, C2
home/work-with-me.md	5	P1, P2, C2
cyber-environments-and-benchmarks/cyber-battle-field.md	3	P1
SUMMARY.md	3	P1, P2
publications.md	2	P2, C3
personal/overview.md	2	C3
cyber-environments-and-benchmarks/cyber-wheel.md	2	P1
toward-deployment/next.md	2	C2
toward-deployment/README.md	2	P2
tools/tool-roadmap-and-status-board.md	1	AMBIGUOUS
resources/research/think.md	1	C3
resources/research/project-management.md	1	C3
resources/career/README.md	1	C3
arl-baa-0122/30-method-candidates-and-metrics.md	1	P6
design-the-game/20-mental-operations-consensus.md	1	C3
be-realistic/sim2sim2real-track.md	1	P2
be-realistic/overview.md	1	P2
cyber-human-ai-dictionary/transfer-learning-terms.md	1	P4
cyber-environments-and-benchmarks/README.md	1	C2
problem-solving/capture-the-flag.md	1	C3
mental-world-modeling/README.md	1	C3
overview-4/micro-opportunities.md	1	H5
mentor/2025-2026-utep.md	1	P1
home/what-is-she-up-to.md	1	P2
home/what-is-she-thinking.md	1	AMBIGUOUS
B. FOE-Dreamer manuscript, 1 file, 11 tokens		
[May26th]_Full/main.tex	11	P2, H1
C. RAISE sections, 9 files, 112 tokens		
6_experimentalresults.tex	51	P1
1_introduction.tex	18	P1, P2
3_methodology.tex	11	P1
2_relatedwork.tex	11	P1, P2
4_env.tex	8	P1
5_transfer.tex	5	P1
9_conclusion.tex	4	P1, P2
5_transfer_old.tex	3	P1
7_discussion.tex	1	P1
D. Learn Structure sections, 5 files, 20 tokens		
taxonomy_v2.tex	11	H2, C1
adapting.tex	4	C1, H4
taxonomy_reorganization_redo.tex	2	H2, H3
appendix_landscape.tex	2	C1
taxonomy.tex	1	C1
E. DesignTheGame core idea, 5 files, 8 tokens		
sections/on_mental_operations.tex	3	C3
main.tex	2	C3
sections/stream-of-thoughts-notes.tex	1	C3
sections/part1_questions_students.tex	1	C3
sections/part1_operations.tex	1	C3
F. PickYourBattles mind map, 8 files, 48 tokens		
questions_clustered_by_topic.md	21	P4, P5, P6, C1, H5, AMBIGUOUS
world_model_taxonomy.md	6	P4, P6

File	Tokens	Senses
world_model_challenges.md	6	P2, P4
psro_lit_notes.md	6	P5, AMBIGUOUS
raw_notes_psro_lit.md	3	P5
opponent_modeling_extraction.md	2	P3, P4
mindmap_draft.md	2	P2, P5
cross_project_model_requirements.md	2	P2, P6
G. August rebuttal, 5 files, 19 tokens		
[Aug25th]_Rebuttal/main.tex	12	P3
v2.tex	2	P3
v1.tex	2	P3
q_n_a.tex	2	P3, P4
analysis.tex	1	P3

Two counts fall out of the table and both are worth keeping. Group I holds 214 of the 267 tokens, and 112 of those sit in one paper. Group IV holds none.

4 The blind pass

The table above has 84 file-and-sense cells. Four were marked AMBIGUOUS on the first pass. The assignment was then run a second time with the sense identifiers relabeled and the first pass hidden. 77 of the 84 cells came back the same, an agreement of 92 percent, and the seven that disagreed are recorded below rather than argued to a verdict. All seven are now AMBIGUOUS, which brings the total to eleven.

1. `training-in-realistic-environments.md`, C2 against P2. The sentence is *which respects it got right is what decides whether a result transfers*. The subject is a result, which argues C2, but the paragraph around it is about policies. Settled by asking whether the sentence stays true when the policy is held fixed and only the claim about it changes.
2. `home/work-with-me.md`, C2 against P1. *Claims about transfer are only meaningful if the same agent can actually be run across the environments being compared*. One clause is about claims and the other is about running an agent, and the sentence exists to link them.
3. `arl-baa-0122`, P6 against P1. *Poor transfer* in the failure column, *unseen-configuration performance retention* in the metric column. The two columns name different senses on one row. Settled by whichever column the experiment is actually built from.
4. `taxonomy_v2.tex`, C1 against C2. *it transfers as a caution*. What moves is a warning from the interpretability debate, which is C1 by object and C2 by function.
5. `world_model_challenges.md`, P2 against P4. *offline-to-real transfer*. Offline data is not a simulator, so the source is a dataset rather than an environment, which pulls it toward P4.
6. `psro_lit_notes.md`, P5 against P4. Whether a later PSRO epoch is a different task. If the opponent distribution is part of the task, the epochs are different tasks and NeuPL is P4. If it is part of the environment, they are not.
7. `cross_project_model_requirements.md`, P6 against P1. *if model trained on one graph size can transfer*. A different graph size is a different configuration, which is P6, unless the per-device

modules have to be recomposed to fit it, which starts to look like building a translator.

The pattern across all seven is one boundary: P1 against P6, and its relatives. Whether the target is a new configuration of one environment or a new environment is the distinction the corpus keeps failing to mark, and it is the distinction that decides whether a wrapper has to be written.

5 Where two of your files disagree

One paper, two senses

[May26th]_Full/main.tex uses the word eleven times. Eight are P2, the policy moving from simulator to network. Three are H1, bytes moving between hosts, and they sit in the attack narrative and the user-activity description where an attacker *dumps tables for outbound transfer* and GHOSTS personas perform *file transfers between PCs*. The two senses are three pages apart in a paper whose contribution is about the first.

One thread, two objects

On the Toward Deployment thread, `training-in-realistic-environments.md` is about a *defensive* policy and argues for not transferring at all, while `transfer-to-realistic-environments.md` is about an *offensive* policy and is the study of how to transfer. The word carries a different object in each, and the two pages sit under one heading. Any lexicon that treats the thread as one usage will merge an argument against a step with a study of that step.

Two papers, two things that vary

In the rebuttal the word means across attacker classes, environment fixed. In the RAISE paper it means across environments, adversary not modeled. Corpus G is entirely P3 and corpus C is entirely P1 and P2, and no file in either uses the other's sense. A reader who has seen one of your papers will supply the wrong default when reading the other.

Same folder, not the same word

`taxonomy_v2.tex` and `adapting.tex` sit beside each other and between them carry four senses: transfer entropy, transferable utility, state transfer in the control sense, and the transfer of a measure out of its home field. Only the last is the word this entry is about. A search over that folder for the term returns mostly noise, which is worth knowing before running one.

6 What to say instead

Six replacements, each pinned to the sense it protects.

- **Say policy transfer, not transfer learning**, for the RAISE work. No learning happens in the target: the policy takes no gradient step there. The encoder does see unlabeled target observations, so the pipeline is transductive domain adaptation while the policy is zero-shot, and the two halves want separate names.

- **Say cross-environment or cross-opponent**, whichever is meant, before the word transfer. That single qualifier separates P1 from P3, and those two are the senses your two papers use.
- **Say unseen configuration**, not transfer, for P6. If no wrapper had to be written, it was not a transfer.
- **Say the result holds**, not the result transfers, for C2. It stops a claim about evidence from being read as a claim about engineering.
- **Say transferability only with an object attached**. Bare, it reads as S1 to anyone who came through adversarial machine learning, where the word is the name of an attack.
- **Rename the transfer assumptions checklist** on the sim2sim2real track page. To a governance reader that phrase is S2, and the checklist is about neither insurance nor contract.

One term is missing from your vocabulary altogether. Pan and Yang give a name to the case where the source and target are unrelated and moving knowledge between them hurts the target: negative transfer. Nothing in 267 tokens names it, and the RAISE result where matching the observation shape without matching its distribution leaves the policy emitting only no-ops is exactly that case. The paper’s own phrase for it is that the failure is total, not graceful.

The neighboring entry is the one this word keeps colliding with, and it has its own three senses already written down in Section 5.1 of the rebuttal. Generalization.